



The Moment Generating Function (MGF)

The Moment Generating Function, denoted $\phi_X(t)$ or $M_X(t)$, is a powerful tool in probability theory used to characterize a random variable X and simplify calculations involving its moments and sums with other independent variables.

1. 🌀 Definition and Existence

A. Formal Definition

The MGF of a random variable X is defined as the **expected value** of the transformed function e^{tX} :

$$\phi_X(t) = E[e^{tX}]$$

Applying the Law of the Unconscious Statistician (LOTUS), the definition expands based on whether X is discrete or continuous:

Type of RV	MGF Formula
Discrete	$\phi_X(t) = \sum_x e^{tx} P(X = x)$
Continuous	$\phi_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$

B. Domain and Existence

The MGF is a function of the real variable t .

- **Existence:** The MGF $\phi_X(t)$ **exists** only if the sum or integral above converges (i.e., yields a finite value) for all t in some open interval around zero, specifically for $t \in (-h, h)$ where $h > 0$.
- **Non-existence:** Distributions with "heavy tails" (like the **Cauchy distribution**) do not have a defined MGF because the integral diverges near $t = 0$.

2. ✨ The "Moment Generating" Property

The MGF is named for its ability to generate the raw moments of the random variable X through differentiation.

A. Definition of Moments

The **k -th raw moment** of a random variable X is defined as the expected value of X raised to the power of k :

$$k\text{-th Moment} = E[X^k]$$

Moment	Name
$E[X^1]$	The Mean (First Moment)
$E[X^2]$	The Second Moment
$E[X^k]$	The k -th Moment

B. The Magic Trick

The k -th raw moment is obtained by taking the k -th derivative of the MGF with respect to t and then evaluating the result at $t = 0$:

$$\phi_X^{(k)}(0) = E[X^k]$$

Proof Intuition: This property comes from the Taylor series expansion of e^{tX} :

$$e^{tX} = 1 + tX + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \dots$$

Taking the expectation (using linearity):

$$\phi_X(t) = E[e^{tX}] = 1 + tE[X] + \frac{t^2}{2!}E[X^2] + \frac{t^3}{3!}E[X^3] + \dots$$

The coefficients of $\frac{t^k}{k!}$ in the Taylor expansion of $\phi_X(t)$ are the moments $E[X^k]$. In calculus, the coefficient of $\frac{t^k}{k!}$ is the k -th derivative evaluated at $t = 0$, i.e., $\phi_X^{(k)}(0)$.

C. Finding Mean and Variance

Statistic	Calculation using MGF
Mean $E[X]$	$E[X] = \phi'_X(0)$ (First derivative at $t = 0$)
$E[X^2]$	$E[X^2] = \phi''_X(0)$ (Second derivative at $t = 0$)
Variance $\text{Var}(X)$	$\text{Var}(X) = E[X^2] - (E[X])^2 = \phi''_X(0) - [\phi'_X(0)]^2$

3. 🚀 The Two Superpowers of MGFs

These properties are what make the MGF an indispensable tool for advanced probability.

Superpower 1: Sums of Independent Variables

This property simplifies finding the distribution of a sum of independent random variables, $Z = X_1 + X_2 + \dots + X_n$.

If $X_1, X_2, \dots, X_n$ are **independent** random variables, the MGF of their sum is the **product** of their individual MGFs:

$$\phi_Z(t) = \phi_{X_1}(t) \cdot \phi_{X_2}(t) \cdot \dots \cdot \phi_{X_n}(t)$$

- **Significance:** This replaces the difficult process of **convolution** (which involves complex integrals or sums) with simple algebraic multiplication.

Superpower 2: The Uniqueness Property

This is the most critical feature that connects the MGF back to the distribution itself.

Theorem: If two random variables, X and Y , have the same Moment Generating Function, then they **must** have the same probability distribution.

$$\text{If } \phi_X(t) = \phi_Y(t) \text{ for } t \in (-h, h) \implies X \sim Y$$

- **Significance:** The MGF acts as a **unique fingerprint** for the distribution. If you calculate the MGF of a sum (Superpower 1) and recognize the resulting function as the MGF of a known distribution, you immediately know the distribution of the sum (Superpower 2).

That's right! Seeing the MGF in action is the best way to grasp its power.

Here are the three key examples demonstrating the MGF's core functions: **generating moments** (in both discrete and continuous cases) and **simplifying sums** (the superpower).

🎯 Example 1: Finding Moments (Discrete)

We will use the MGF to find the mean ($E[X]$) and variance ($\text{Var}(X)$) of a **Bernoulli** random variable, X .

The Bernoulli Distribution $Bern(p)$

- **Definition:** $P(X = 1) = p$ and $P(X = 0) = 1 - p$.
- **Goal:** Use the MGF to verify that $E[X] = p$ and $\text{Var}(X) = p(1 - p)$.

1. Find the MGF $\phi(t)$

Using the discrete MGF definition $\phi(t) = \sum e^{tx} P(X = x)$:

$$\phi(t) = (e^{t \cdot 0} \cdot P(X = 0)) + (e^{t \cdot 1} \cdot P(X = 1))$$

$$\phi(t) = (1 \cdot (1 - p)) + (e^t \cdot p)$$

$$\phi(t) = 1 - p + pe^t$$

2. Find the Mean $E[X]$ (1st Moment)

We use $E[X] = \phi'(0)$:

- **First Derivative:**

$$\phi'(t) = \frac{d}{dt}(1 - p + pe^t) = 0 + pe^t = pe^t$$

- **Evaluate at $t = 0$:**

$$E[X] = \phi'(0) = pe^0 = p \cdot 1 = p$$

(Correct, the mean of a single success/failure trial is p).

3. Find the Variance $\text{Var}(X)$

We need $E[X^2] = \phi''(0)$:

- **Second Derivative:**

$$\phi''(t) = \frac{d}{dt}(pe^t) = pe^t$$

- **Evaluate at $t = 0$:**

$$E[X^2] = \phi''(0) = pe^0 = p$$

- **Calculate Variance:** $\text{Var}(X) = E[X^2] - (E[X])^2$

$$\text{Var}(X) = p - (p)^2 = p(1 - p)$$

(Correct, the variance is confirmed using the MGF).

Example 2: Finding Moments (Continuous)

We will find the mean of an **Exponential** random variable, X , which models waiting times.

The Exponential Distribution $\text{Exp}(\lambda)$

- **PDF:** $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.
- **Goal:** Use the MGF to verify that $E[X] = 1/\lambda$.

1. Find the MGF $\phi(t)$

Using the continuous MGF definition $\phi(t) = \int_0^\infty e^{tx} f(x) dx$:

$$\phi(t) = \int_0^\infty e^{tx} (\lambda e^{-\lambda x}) dx$$

$$\phi(t) = \lambda \int_0^{\infty} e^{(t-\lambda)x} dx$$

- **Solve the integral** (assuming $t < \lambda$ for convergence):

$$\phi(t) = \lambda \left[\frac{e^{(t-\lambda)x}}{t-\lambda} \right]_0^{\infty}$$

$$\begin{aligned}\phi(t) &= \lambda \left(0 - \frac{e^0}{t-\lambda} \right) = \lambda \left(\frac{-1}{t-\lambda} \right) = \frac{\lambda}{\lambda-t} \\ \phi(t) &= \lambda(\lambda-t)^{-1}\end{aligned}$$

2. Find the Mean $E[X]$ (1st Moment)

We use $E[X] = \phi'(0)$:

- **First Derivative** (using the Chain Rule):

$$\phi'(t) = \frac{d}{dt}(\lambda(\lambda-t)^{-1})$$

$$\phi'(t) = \lambda \cdot (-1)(\lambda-t)^{-2} \cdot (-1) = \lambda(\lambda-t)^{-2} = \frac{\lambda}{(\lambda-t)^2}$$

- **Evaluate at $t = 0$:**

$$E[X] = \phi'(0) = \frac{\lambda}{(\lambda-0)^2} = \frac{\lambda}{\lambda^2} = 1/\lambda$$

(Correct, the mean waiting time is the inverse of the rate λ).

✨ Example 3: The Superpower (Sums and Uniqueness)

We use the MGF to find the distribution of the sum of two independent **Poisson** random variables, $Z = X + Y$.

The Poisson Distribution $Pois(\lambda)$

- **Given MGF:** The MGF for a single $Poisson(\lambda)$ variable is:

$$\phi_{\lambda}(t) = e^{\lambda(e^t-1)}$$

- **Setup:** Let $X \sim Pois(\lambda_1)$ and $Y \sim Pois(\lambda_2)$, and X and Y are independent.
- **Goal:** Find the distribution of $Z = X + Y$ without using convolution.

1. Use Superpower 1: MGF of the Sum is the Product

$$\phi_Z(t) = \phi_X(t) \cdot \phi_Y(t)$$

$$\phi_Z(t) = \left(e^{\lambda_1(e^t-1)} \right) \cdot \left(e^{\lambda_2(e^t-1)} \right)$$

2. Simplify Algebraically

Using the rule $e^a \cdot e^b = e^{a+b}$:

$$\phi_Z(t) = e^{\lambda_1(e^t-1) + \lambda_2(e^t-1)}$$

Factor out the common term $(e^t - 1)$ from the exponent:

$$\phi_Z(t) = e^{(\lambda_1 + \lambda_2)(e^t - 1)}$$

3. Use Superpower 2: Uniqueness

We compare the resulting MGF, $\phi_Z(t) = e^{(\lambda_1 + \lambda_2)(e^t - 1)}$, to the general form of the Poisson MGF, $\phi_\lambda(t) = e^{\lambda(e^t - 1)}$.

The function $\phi_Z(t)$ is the **exact fingerprint** of a Poisson distribution, but with the new parameter $\lambda = \lambda_1 + \lambda_2$.

Conclusion: By the Uniqueness Property, the sum Z must be a Poisson variable:

$$\mathbf{Z} \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

This transformation from difficult calculus (convolution) to simple algebra (multiplication) is the greatest benefit of the MGF.

That's a fantastic idea! Compiling the MGFs for the core distributions creates a powerful reference table.

Here is the derivation or stated MGF for every distribution we covered previously, organized by discrete and continuous types.

Discrete Probability Distributions

Distribution	Parameters	Probability Mass Function (PMF)	Moment Generating Function $\phi(t)$
Bernoulli	p	$P(X = k) = p^k(1 - p)^{1-k}$	$\phi(t) = 1 - p + pe^t$
Binomial	n, p	$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$	$\phi(t) = (1 - p + pe^t)^n$
Poisson	λ	$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$	$\phi(t) = e^{\lambda(e^t - 1)}$
Geometric	p	$P(X = k) = (1 - p)^{k-1} p$	$\phi(t) = \frac{pe^t}{1 - (1-p)e^t}$
Hypergeometric	N, K, n	$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$	Does Not Exist in a Simple Closed Form

Derivation Notes for Discrete MGFs

1. **Bernoulli** $Bern(p)$: (Derived in the previous example)

$$\phi(t) = \sum_{k=0}^1 e^{tk} P(X = k) = e^{t \cdot 0}(1 - p) + e^{t \cdot 1}p = 1 - p + pe^t$$

2. **Binomial** $Bin(n, p)$: A Binomial variable is the sum of n independent Bernoulli variables ($X = X_1 + \dots + X_n$). Using the MGF Superpower 1 (product of MGFs):

$$\phi_X(t) = \phi_{X_1}(t) \cdot \dots \cdot \phi_{X_n}(t) = (1 - p + pe^t) \cdot \dots \cdot (1 - p + pe^t)$$

$$\phi_X(t) = (1 - p + pe^t)^n$$

3. **Poisson** $Pois(\lambda)$:

$$\phi(t) = \sum_{k=0}^{\infty} e^{tk} P(X = k) = \sum_{k=0}^{\infty} e^{tk} \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!}$$

Recognizing the Taylor series for e^z : $\sum_{k=0}^{\infty} \frac{z^k}{k!} = e^z$. Let $z = \lambda e^t$.

$$\phi(t) = e^{-\lambda} e^{\lambda e^t} = e^{\lambda e^t - \lambda} = e^{\lambda(e^t - 1)}$$

4. **Geometric** $Geom(p)$:

$$\phi(t) = \sum_{k=1}^{\infty} e^{tk} P(X = k) = \sum_{k=1}^{\infty} e^{tk} (1-p)^{k-1} p$$

$$\phi(t) = pe^t \sum_{k=1}^{\infty} (e^t(1-p))^{k-1}$$

This is a geometric series $\sum_{j=0}^{\infty} r^j = \frac{1}{1-r}$, where $r = e^t(1-p)$ and $j = k-1$.

$$\phi(t) = pe^t \left(\frac{1}{1 - e^t(1-p)} \right) = \frac{pe^t}{1 - (1-p)e^t}$$

5. **Hypergeometric:** Due to the dependence between trials (sampling without replacement), the MGF is very complex and is rarely used.

Continuous Probability Distributions

Distribution	Parameters	PDF $f(x)$	Moment Generating Function $\phi(t)$
Uniform	a, b	$\frac{1}{b-a}$ for $a \leq x \leq b$	$\phi(t) = \frac{e^{tb} - e^{ta}}{t(b-a)}$
Exponential	λ	$\lambda e^{-\lambda x}$ for $x \geq 0$	$\phi(t) = \frac{\lambda}{\lambda - t}$
Normal (General)	μ, σ^2	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$\phi(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$
Standard Normal	$\mu = 0, \sigma^2 = 1$	$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$	$\phi(t) = e^{\frac{1}{2}t^2}$
Chi-Squared	k (dof)	- (Derived from Normal)	$\phi(t) = (1 - 2t)^{-k/2}$

Derivation Notes for Continuous MGFs

1. **Exponential $Exp(\lambda)$:** (Derived in the previous example)

$$\phi(t) = \frac{\lambda}{\lambda - t}$$

2. **Uniform $U(a, b)$:**

$$\phi(t) = \int_a^b e^{tx} \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{e^{tx}}{t} \right]_a^b$$

$$\phi(t) = \frac{1}{t(b-a)} (e^{tb} - e^{ta})$$

3. **Normal $N(\mu, \sigma^2)$:** The derivation requires completing the square in the exponent of the integral and is algebraically intensive.

$$\phi(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

• **Standard Normal $N(0, 1)$:** Set $\mu = 0$ and $\sigma^2 = 1$:

$$\phi(t) = e^{0 \cdot t + \frac{1}{2} \cdot 1 \cdot t^2} = e^{\frac{1}{2}t^2}$$

4. **Chi-Squared $\chi^2(k)$:** A Chi-Squared variable with k degrees of freedom is the sum of k independent Standard Normal variables squared:

$$X = Z_1^2 + \dots + Z_k^2.$$

- The MGF of a single Z^2 (where $Z \sim N(0, 1)$) can be calculated to be $(1 - 2t)^{-1/2}$.
- Using the sum property, the MGF of X is the product of k such MGFs:

$$\phi_X(t) = \phi_{Z_1^2}(t) \cdot \dots \cdot \phi_{Z_k^2}(t) = \left((1 - 2t)^{-1/2} \right)^k$$

$$\phi_X(t) = (1 - 2t)^{-k/2}$$
